Class XII Maths Sample Paper – Set 5

Time Allowed: 30 minutes

General Instructions:

1. This Question paper contains - five sections A, Each section is compulsory.

However, there are internal choices in some questions.

2. Section A has 18 MCQ's and 02 Assertion-Reason based questions of 1 mark each.

Q. No.	Questions	Marks
	Section A	
1.	If $A=[a_{ij}]$ is a scalar matrix of order $n\times n$ such that $a_{ij}=k$ for all i, then trace of A is equal to (a) n - k (b) nk	1

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	(c) $n+k$ (d) n^k	
2.	Let A and B be two invertible matrices of order 3×3. If $\det(ABA')=8$ and $\det(AB^{-1})=8$, then $\det(BA^{-1}B')$ is equal to (a) -2 (b) 8 (c) 2 (d) -8	1
3.	3. If A is a 3×3 matrix and $\lvert A\rvert=-2$, then value of $\lvert A(\text{adj}A)\rvert$ is (a) cot e^x (b) $e^x \cot(e^x)$ (c) cosec e^x (d) $e^x \csc(e^x)$	1
4.	4. If $y=\log(\sin(e^x))$, then $\frac{dy}{dx}$ is:	1

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	(a) $\cot(e^x)$ (b) $e^x \cot(e^x)$ (c) $\csc(e^x)$ (d) $e^x \csc(e^x)$	
5.	The direction ratios of a line parallel to z -axis are: (a) $\langle 0,0,0 \rangle$ (b) $\langle 1,1,0 \rangle$ (c) $\langle 1,1,1 \rangle$ (d) $\langle 0,0,1 \rangle$	1
6.	What is integrating factor of $\frac{dy}{dx} + y\sec x = \tan x$? (a) $\sec x$ (b) $e^{\sec x}$ (c) $\sec x + \tan x$ (d) $\log(\sec x + \tan x)$	1
7.	The corner points of the feasible region determined by the system of linear constraints are $(0,10)$, $(5,5)$, $(15,15)$, $(0,20)$. Let $Z = px + qy$,	1

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7.	where $p, q > 0$. Condition on p and q so that the maximum of Z occurs at both the points $(15,15)$ and $(0,20)$ is (a) $q = 3p$ (b) $q = 2p$ (c) $p = q$ (d) $p = 2q$	
8.	Which of the following is the principal value branch of $\csc^{-1} x$? (a) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ (b) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ (c) $[0,\pi](\frac{\pi}{2})$ (d) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$	1
9.	$\int_0^{\frac{\pi}{4}} \frac{e^{\tan x}}{\cos^2 x} dx =$	1

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	(a) $e-1$ (b) $\frac{1}{e}-1$ (c) $e+1$ (d) $\frac{1}{e}+1$	
10.	The order of the single matrix obtained from $\begin{pmatrix} 1 & 0 \\ 2 & -1 \\ 2 & 3 \end{pmatrix}_{3\times 2} \begin{pmatrix} -1 & 2 & 0 \\ 0 & 2 & 1 \end{pmatrix}_{2\times 3} \begin{pmatrix} 0 & 1 & 1 \\ 0 & \frac{2}{3} & 2 \end{pmatrix}_{2\times 3}$ is (a) 2×3 (b) 3×3 (c) 3×2 (d) 2×2	1
11.	The solution set of the inequality $3x+5y<4$ is (a) an open half-plane not containing the origin (b) an open half-plane containing the origin (c) a closed half plane containing the origin (d) the whole XY-plane not containing the line $3x+5y=4$.	1

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12.	If the vectors $\langle 4,1,1\rangle$ and $\langle 7,2,6\rangle + m\langle 5,4,1\rangle$ are coplanar, then m is (a) 16 (b) -10 (c) 0 (d) None of these	1
13.	$\begin{vmatrix} a+ib & -c+id \\ c+id & a-ib \end{vmatrix} = ?$ (a) $(a^2-b^2-c^2-d^2)$ (b) $(a^2+b^2+c^2-d^2)$ (c) $(a^2-b^2+c^2-d^2)$ (d) $(a^2+b^2+c^2+d^2)$	1
14.	If $P(A)=\frac{3}{10}, P(B)=\frac{2}{5}$ and $P(A\cup B)=\frac{3}{5}$, then $P(B/A)+P(A/B)$ equals (a) $\frac{1}{3}$ $(b)\ \frac{1}{4}$ $(c)\ \frac{5}{12}$ $(d)\ \frac{7}{12}$	1

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15.	The solution of the differential equation $\frac{dy}{dx} = \frac{e_x - y}{x^2 e^{-y}}$ is (a) $e^x + e^y = \frac{x^3}{3} + C$ (b) $e^x - e^y = \frac{x^3}{3} + C$ (c) $y = e^{x-y} - x^2 e^{-y} + C$ (d) $e^y - e^x = \frac{x^3}{3} + C$	1
16.	The area of a triangle with vertices A,B,C is given by (a) $\lvert \overrightarrow{AB} \times \overrightarrow{AC} \rvert$ (b) $\frac{1}{8} \lvert \overrightarrow{AC} \times \overrightarrow{AB} \rvert$ (c) $\frac{1}{4} \lvert \overrightarrow{AC} \times \overrightarrow{AB} \rvert$ (d) $\frac{1}{2} \lvert \overrightarrow{AB} \times \overrightarrow{AC} \rvert$	1
17.	If $f(x) = \begin{cases} kx+5, & x \le 2, \\ x+1, & x > 2 \end{cases}$ is continuous at $x = 2$ then $k =$?	1

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	(a) -2 (b)-1 (c)2 (d)-3	
18.	The lines l_1 and l_2 intersect. The shortest distance between them is (a) infinity (b)negative (c)positive (d)0	1
19.	Assertion (A): The rate of change of area of a circle with respect to its radius r when $r = 6$ cm is 12π cm^2/cm. Reason (R): Rate of change of area of a circle with respect to its radius r is $\frac{dA}{dr}$, where A is the area of the circle. (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true.	1

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20.	Assertion (A): A relation $R=\{(a,b):\lvert a-b\rvert<3\}$ defined on the set $A=\{1,2,3,4\}$ is reflexive. Reason (R): A relation R on the set A is said to be reflexive if for $(a,b)\in R$ and $(b,c)\in R$, we have $(a,c)\in R$. (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is not the correct explanation of A. c) A is true but R is false. (d) A is false but R is true.	1

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